INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP
Project Presentation

Intelligent Voice Controlled Communication Assistant

Secure • Voice-first • Multi-channel

VoiceMail AI

Hands-free email & messaging with smart summaries and secure actions

Email Reader
Read, search, summarize

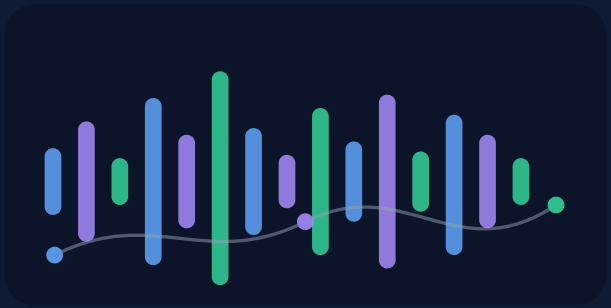
Messaging
WhatsApp & Telegram

Secure Actions
OAuth + Voice PIN

Developer
Himanshu Srivastava

Deployment-ready • Docker • Render

VOICE SESSION
Listening...



Example command
"Read my latest email"

Action
Summarize + Reply

Voice → Intent → Services → Response • Online

Modern AI / technology themed design • Clean visuals • Minimal text

1280×720 • VoiceMail AI

VOICEMAIL AI
Project Overview

Voice-first productivity

A voice-controlled assistant for email & messaging

VoiceMail AI listens, understands intents, and performs secure actions across communication channels—
so users can read, reply, and send messages hands-free.

LIVE SESSION
Listening & Executing

Recognized intent
Read latest email + summarize

Ready

Example command
“Summarize unread emails”

Example command
“Send WhatsApp to Rahul”

Reads & summarizes emails
Fetches messages, reads them aloud, and highlights key points for quick triage.

Sends messages hands-free
Sends updates across channels (email, WhatsApp, Telegram) without typing.

Understands voice commands
Detects intent (read/search/reply/send) and routes actions via the backend.

Fast, secure, accessible

VOICEMAIL AI
Milestone 1

OAuth 2.0 + Secure Sessions

Milestone 1: User Authentication

A secure sign-in foundation enabling safe access to Gmail and messaging actions with session control and role-based permissions.

MILESTONE
01

Least-privilege access

AUTH FLOW

Google OAuth → Backend Session

1) Google consent & sign-in — User
OAuth 2.0 authorization code flow with requested scopes.

2) Flask callback & token exchange — Backend
Server validates state and creates a secure session/JWT.

3) Store user + refresh token — DB
SQLite user store for sessions, refresh, and audit-ready metadata.

4) Role-based access & logout — Policy
Protect sensitive actions with RBAC and session termination.

Google OAuth 2.0 sign-in

Consent-based authentication for accessing Gmail features securely.

Flask backend sessions / JWT

Secure session handling with protected routes for voice-triggered actions.

SQLite user store + refresh

Lightweight persistence for user profiles, tokens, and session metadata.

VOICEMAIL AI
Milestone 2

SpeechRecognition + Web Speech API

Milestone 2: Voice Command Engine

Hands-free interaction layer that converts speech to intent, executes actions, and speaks back confirmations with robust noise handling.

MILESTONE
02

Low-latency voice UX

VOICE PIPELINE

Speech → Intent → Action → TTS

1) Capture audio input (ASR)
SpeechRecognition / Web Speech API converts voice to text.
ASR

2) Parse intent & route command
Detect action (read, search, send) and required parameters.
NLP

3) Execute service action
Calls backend handlers for Gmail / WhatsApp / Telegram tasks.
Exec

4) Speak back response (TTS)
Text-to-Speech generates confirmations and guided prompts
TTS

Speech-to-Text (ASR)

Converts user speech into text using SpeechRecognition and browser Web Speech API where available.

Intent parsing & routing

Maps utterances to intents and parameters (recipient, subject, message) for reliable command execution.

Text-to-Speech feedback

Reads emails aloud and confirms actions with clear spoken prompts for accessibility and speed.

VOICEMAIL AI
Milestone 3

Gmail • WhatsApp • Telegram

Milestone 3: Messaging & Email Integrations

Connected core channels through secure APIs to fetch, send, and automate communications from voice commands.

MILESTONE
03

Multi-channel routing

INTEGRATION LAYER

Service connectors used by the intent router

Gmail API
Email fetch + send
OAuth

- Read/unread listing & thread details
- Search by sender/subject/keywords
- Compose, send, and manage labels

WhatsApp
via Twilio API
Twilio

- Send messages to contacts/groups
- Inbound webhooks for delivery updates
- Templated alerts & confirmations

Telegram
Bot API
Bot

- Command-driven workflows (/read, /send)
- Secure notifications and message delivery
- Webhook polling support for updates

ROUTING

Intent → Connector

Command parsing — ASR text
Detect channel + action (read / send / notify).

Connector selection — Gmail/Twilio/Telegram
Route to the correct API adapter with auth context.

Confirmation — TTS + UI
Speak back status and log delivery results.

VOICEMAIL AI
AI Features

Summarize • Suggest • Verify

Intelligent assistance on top of voice commands

VoiceMail AI applies lightweight AI to reduce reading time, speed up replies, and add a safety layer before executing sensitive actions.

FEATURE SUITE

Three AI capabilities used in everyday workflows

Email Summarization
AI
Quick triage

- Extracts key points and action items
- Works well for long threads & updates
- Read aloud with concise voice output

Suggested Replies
Assist
Faster responses

- Drafts short replies using context
- Tone options: formal / friendly / brief
- User confirms before sending

Voice PIN Verification
Secure
Safety layer

- Required before sending/critical actions
- Prevents accidental or unauthorized commands
- Logs verification attempts for audit

AI PIPELINE

Where intelligence is applied

1) Context — Email / Thread
Extract key text + metadata for decision making.

2) Generate — Summary / Reply
Create concise outputs optimized for speech.

3) Guardrails — Voice PIN
Verify before sending messages or changing state.

SAMPLE OUTPUT

VOICEMAIL AI
Milestone 4

Analytics + Admin Controls

Milestone 4: Admin Dashboard & Analytics

MILESTONE
04

Visibility and control across users, commands, and delivery outcomes—supporting monitoring, auditing, and continuous improvement.

Export-ready reporting

DASHBOARD VIEW

Usage, Delivery & Errors at a Glance

Active users (7d)
Tracked via sessions

Command volume
By intent/category

Delivery success
Email & messages

Error rate
API & ASR issues

Command Usage (sample view)
Compare top intents to spot adoption patterns.

Insights

Read / Search Email — High

User activity tracking

Session logs and event history to support auditing and troubleshooting.

Command usage metrics

Breakdown by intent, channel, and frequency to guide UX improvements.

Delivery success & error rates

Monitor API calls and message delivery outcomes with actionable error logs.

VOICEMAIL AI
System Architecture

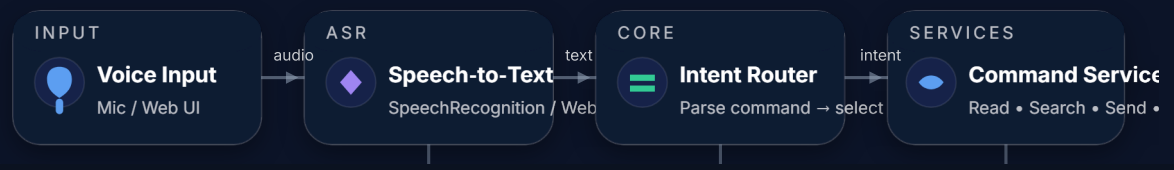
Voice → Backend → Integrations → Response

End-to-end flow from speech to secure action

VoiceMail AI converts voice input into intent, executes channel-specific services through authenticated APIs, and returns confirmations via speech and UI.

ARCHITECTURE FLOW

Voice pipeline with secure routing



CORE MODULES

What runs where

Client — Browser / mic
Captures speech and displays confirmations.

Backend — Flask
Intent routing, auth checks, connector orchestration.

Storage — SQLite
User profile, token metadata, usage logs.

Connectors — Gmail / Twilio / Telegram
Unified adapter interface for multi-channel

VOICEMAIL AI
Technology Stack

Backend • Voice • APIs • Deployment

Tech stack built for secure, voice-first automation

A lightweight Python + Flask backend, voice interfaces for ASR/TTS, and authenticated connectors for Gmail, WhatsApp, and Telegram—containerized for deployment.

CATEGORIES

Core components and libraries

Backend
API + session logic

Core

Python | Flask

Role
Routing, auth checks, webhooks

Data
Users, tokens, logs

Local

SQLite

Role
Persistence + audit-ready metadata

DEPLOYMENT

Containerized release

Docker
Build

Reproducible environment for local dev and production.

Render
Host

Simple web service deployment with logs and health checks.

Operational practice
Secrets kept outside source code

VOICEMAIL AI
Outcome & Impact

Productivity • Accessibility • Secure automation

Delivering hands-free communication that's fast, secure, and scalable

VoiceMail AI demonstrates an end-to-end voice workflow—authenticate, understand intent, execute actions across channels, and confirm outcomes—while keeping user control at the center.

IMPACT SUMMARY

What the project achieves

Hands-free productivity

Voice UX

Voice-first actions reduce context switching—read, search, and respond while multitasking.

Faster inbox triage

AI assist

Summaries highlight key points and actions—optimized for speech output.

NEXT

Roadmap enhancements

Calendar integration

Productivity

Schedule meetings, read agendas, and create reminders by voice.

Multilingual support

Access

Multi-language ASR/TTS to serve diverse users and improve accessibility.

Offline mode

Resilience

Local command handling and queued actions for

VOICEMAIL AI
Closing

Demo-ready • Voice-first • Secure by design

Thank You

VoiceMail AI showcases an end-to-end voice workflow: authenticate → understand intent → act across channels → confirm securely.

Hands-free UX
ASR + TTS loop

Secure actions
OAuth + confirmations

Multi-channel
Gmail • WhatsApp • Telegram

DEVELOPER
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Email Contact
{Fill in email address}

LinkedIn Profile
{Fill in profile link}

GitHub Code
{Fill in repository link}

ACKNOWLEDGMENT
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Virtual Internship
Built as part of the Infosys Springboard Virtual Internship learning program.